

AI-Based Indian Sign Languages to Speech Converter Using MediaPipe and Machine Learning

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Abstract—In current scenario, there is a big communication gap between people who are hard of hearing and the rest of society. While many hearing-impaired individuals use Indian Sign Language (ISL) to communicate, most other people rely on spoken or written words. This difference makes difficulty in daily interaction. When people from both communities try to talk to each other in places like schools, hospitals, and public spaces, they face a lot of difficulties. On top of that, there are not enough sign language interpreters available, and helpful technology is often too expensive, making it hard for people to have live, face-to-face conversations. This paper introduces a system that easily converts sign language into text and speech in real time. Using a basic webcam, machine learning, and MediaPipe technology, the system tracks hand movements and translates them into both English and Odia. The process works in a few simple steps: a webcam captures the live video, MediaPipe Hands recognizes the specific gestures, the system writes out the correct text in both languages, and text-to-speech engines read that text out loud. We trained the model using a unique dataset of 720 images of sign language gestures, divided into 35 classes obtained from publicly available repositories. The experimental outcomes showed an accuracy rate of nearly 99% for the system and relatively low delay times, supporting the feasibility of its use in real-time. This system offers a economical, scalable and effective means of providing assistive communications between those who are hearing impaired and the general population and supports multilingual interactions.

Index Terms—Indian Sign Language Recognition, MediaPipe for Hand landmark extraction, Random Forest Classifier, Computer Vision, Machine Learning techniques, Multilingual Text Conversion, English and Odia Speech Synthesis, Real-Time Gesture Recognition Systems.

I. INTRODUCTION

Communication is a vital part of everyday life; however, it is tough for people who cannot hear or speak to communicate easily because they depend heavily on sign language to get their message across. Millions of people across the world use different types of sign languages as their main way of expressing themselves. Despite big steps forward in technology, there is still a massive difficulties between sign language users and normal users.

In the past, communication support services relied on human interpreters or hardware-based tools, including sensor gloves or other wearable devices. This has made it extremely difficult to have low-cost, easy-to-maintain solutions for effective communication on a regular, day-to-day basis. With the rapid growth of artificial intelligence, computer vision, and machine

learning, camera-based sign language recognition systems have become available at much lower costs, making them much more efficient and scalable for real-time use.

The proposed system focuses on recognizing Indian Sign Language (ISL) in real time. It uses MediaPipe's hands tool and the Random Forest classification method to recognize and turn a user's hand gestures, captured by a webcam, into meaningful outputs. This system gives sign language users the ability to turn the recognized gestures into both English and Odia text at the same time. Unlike many current systems, which usually only turn ISL gestures into one spoken language, this model supports turning gestures into both English and Odia language outputs.

Deep learning models like CNN, Transformer and pose estimation techniques are used by most current sign language recognition methods. These techniques typically take substantial computational resources, need large quantity of data, and require high powered hardware; making real-time implementation of them impractical. The proposed system, however, utilizes a lightweight and efficient machine learning model to deliver sufficient accuracy utilizing low latency and thus is more applicable for practical real-time communications and assistive technologies.

This paper developed a Real-time Lightweight AI-Based Sign Language Recognition and Speech Conversion System using Media Pipe land marking Whiteboard and Machine Learning techniques. The core of our system is to process real-time gestures using captures from a Webcam, extract their hand landmarks through Media Pipe Hands, and classify them with a random forest classifier, as opposed to processing complete images. Using landmarks reduces the complexity of processing by a large amount and will still yield a high accuracy of recognition.

To test and develop the proposed system we captured real-time hand gestures via webcam, extracted the hand landmarks using Media Pipe, and classified the Alphanumerical gestures: (A-Z) and (0-9) using the Random Forest Classifier. When a gesture is identified as recognized via the classifier, it is displayed as the relevant character or number on the screen in both the English and Odia version of the text. The recognized text characters are then generated into English and Odia speech audio, therefore providing a more effective method of communication for users between the hearing impaired and a

wider audience.

The proposed Real time AI based Sign Language Recognition/Conversion System will operate on traditional CPU computers as opposed to needing a dedicated GPU for processing, making it affordable and reliable for Real time applications. The combination of gesture recognition, Multi language text generation, and Multi language speech synthesis creates an intelligent and accessible assistive communication medium between hearing-impaired individuals and the larger community.

The major part of this paper are :

- Development Of A Real-time Indian Sign Language (ISL) Recognition System Using MediaPipe Hand Landmarks.
- Implementation Of A Random Forest Classifier To Accurately Recognize ISL Gestures.
- Generating Multilingual Text Outputs (Odia English) To Correspond With ISL Gestures
- Implementing Bilingual Text-to-Speech (TTS) Technologies To Create Separate Speech Syntheses For Odia And English Languages
- Developing An Affordable, Scalable, And Real-time Assistive Communication System That Will Run Efficiently On Standard CPU Devices.

The rest of this paper is organized as follows. Section II reviews past work in this field, Section III explains our approach and methods, Section IV details how the dataset was collected and cleaned, Section V shows the results and explains how the system was tested, and Section VI wraps up the paper and suggests ideas for future research.

II. LITERATURE REVIEW

In the past few years, sign language recognition has become a hot topic because of improvements in artificial intelligence (AI) and computer vision and Machine Learning. Many researchers are working on different ways to recognize gestures/sign language as well as translate it into text using either computer vision or AI technologies.

One of the first end-to-end translation systems for translating sign language into English was developed by **Necati Cihan Camgoz et al** in 2018 . This system was built using a neural network architecture trained with the RWTH-PHOENIX-Weather-2014T dataset that are directly translate sign language videos into English sentence. However, it was limited to being appropriate for large scale datasets and also having a large amount of computing power, as well as limited usability in terms of lighter-weight and real-time implementation on lower-end hardware.

The following version of their system included Sign Language Transformers, which used a transformer network architecture to provide joint recognition of sign language and translation. In this method, the overall accuracy of the translations improved due to better semantic understanding of the gestures as well as elimination of the need for manual alignment of the gesture to text. Nevertheless, transformer networks remain computationally intensive and therefore challenging to

implement for real-time applications. Moreover, the memory consumption required to train a transformer architecture, as well as the length of training time, are both extremely high compared to other models.

Multi-channel transformer architectures were also proposed that would allow for simultaneous collection of data from three different sources to capture individual features of the hands, face, and body motion. Although these multi-channel architectures increased the ability of the model to understand the overall context of each feature, preprocessing and overall training of a model with three different data sources created a great deal of increased complexity compared to prior methods.

A novel that represent the learning for sign language was introduced by **SignBERT**, which utilized self-supervised based tokens to create representations of sign language. The model has been able to successfully perform signer-independent representation learning, also improve generalization of the learned representations when there are fewer available training examples. However, the method included complicated pre-training stages that made it challenging to implement. Additionally, **SignBERT** has required significant computation resources and longer training times than the other methods.

The **JRRS** gesture recognition system used custom gesture dataset for the implementation of a gesture recognition system that is deployed in real-time for practical use. The **JRRS** system demonstrated real-time gesture detection and the ability to convert gestures into text, but the performance of the system decreased due to the limited amounts of data used to train and test the gesture recognition system. Also, under varying light and background conditions, the performance of the gesture recognition system was reduced.

By using deep learning to translate ASL into speech, **Paneru et al's** proposed ASL to speech conversion system achieved a significant proportion of accurate gesture recognition. However, the model required specialized hardware and enormous computational requirements thus causing difficulty in the affordability and convenience of the system for everyday users.

Table I summarizes existing research works.

While there have been notable advancements to date, there are still obstacles faced by existing systems including hardware dependencies, large-sized datasets, computationally intensive workloads and difficulties in implementing systems for use in real time. The design of this system overcomes these challenges with the introduction of a lightweight landmark based machine learning method.‘

III. PROPOSED METHODOLOGY

A. System Overview

The process flow of the implemented system is presented below in Figure ??.

TABLE I
COMPARISON OF EXISTING SIGN LANGUAGE RECOGNITION TECHNIQUES

Technique	Year	Purpose	Advantages	Limitations
Neural Sign Language Translation	2018	End-to-end sign translation	Benchmark framework	Low BLEU accuracy
Sign Transformers	2020	Joint recognition and translation	Improved semantic understanding	High computational complexity
Multi-Channel Transformer	2020	Multi-modal recognition	Better contextual learning	Expensive training
SignBERT	2021	Self-supervised representation learning	Signer-independent learning	Complex pipeline
JRRS System	2019	Real-time gesture recognition	Practical implementation	Limited dataset
ASL to Speech Translation	2024	Sign-to-speech conversion	High accuracy	Hardware dependency

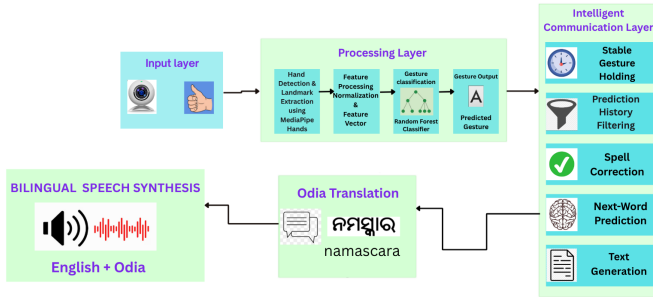


Fig. 1. Proposed System Architecture

The system that we developed, records the hand gestures via webcam and extracts landmark coordinates using MediaPipe Hands. These coordinates are preprocessed into features and classified using a Random Forest Classifier to identify the captured gestures. Text results are constructed both for the English and Odia language from the recognized gestures. Text generated in both English and Odia can utilize text to speech technology to convert text into separate English speech and Odia speech. It also provides real-time and effective communication assistance.

B. Gesture Capture

We use OpenCV to capture real-time video from a standard webcam. This will allow for ongoing monitoring and processing of hand movements. The video frames captured will be continuously processed for detecting and identifying sign language hand gestures with low latency. Therefore, by utilizing a webcam-based input device, this system does not require expensive hardware devices such as sensor gloves or other wearable gear; thus, this will make the solution portable, and easily implemented on any standard CPU-based computing system.

C. MediaPipe Landmark Extraction

By using MediaPipe Hands, the system easily detects hand landmarks in real-time from a standard webcam stream. It

accurately detects 21 specific points across the palm, joints, and fingertips, providing an exact digital representation of hand movements and gestures.

To represent each detected landmark, it uses two representation values (coordinate x, y) respectively:

$$L_i = (x_i, y_i) \quad (1)$$

where L_i denotes the coordinate position of the i^{th} hand landmark, while x_i and y_i represent the horizontal and vertical spatial coordinates, respectively.

The extracted landmarks collectively represent the spatial configuration of the hand gesture.

D. Feature Vector Generation

In order to classify gestures, the extracted hand landmarks are converted to numerical feature vectors. Because MediaPipe detects 21 hand landmarks, each consisting of an (x, y) coordinate—the resulting feature vector comprises 42 distinct numerical attributes. The input feature vector will be preprocessed through normalization techniques that reduce variations introduced through hand orientation, scale, and distance from the camera so that consistent gesture classification can be performed.

E. Gesture Classification

This gesture recognition system utilizes a Random Forest Classifier model. This algorithm implements an Ensemble learning method using a collection of Decision Trees to achieve a more accurate and stable classification and prediction model. During classification, the individual decision trees vote on the gesture, and the system assigns the final gesture class based on whichever category secures the majority of these votes.

$$C(x) = \text{mode}\{T_1(x), T_2(x), \dots, T_n(x)\} \quad (2)$$

where $T_i(x)$ represents prediction of the i^{th} decision tree. The Random Forest classifier was selected because of its:

- High classification accuracy
- Fast training and prediction time
- Robustness against overfitting
- Low computational complexity

- Real-time prediction capability

While alternative deep learning architectures like CNNs, LSTMs, and Transformers were considered, their heavy training requirements and computational resource demands outweigh those of the Random Forest classifier, making them less ideal for resource-constrained or low-latency environments.

F. Text and Speech Generation

Once the system identifies a gesture, it displays the predicted label as text on the user interface. Then this text passes through a SymSpell-driven auto-correction module to improve sentence structure and accuracy. Next, a translation module converts the corrected text into English and Odia, then Google TTS translates both versions into clear, audible speech. This integrated module facilitates efficient real-time communication between hearing impaired individuals and non sign language users through multilingual text and speech outputs.

The speech output enables communication between hearing-impaired individuals and non-sign-language users.

IV. DATASET AND PREPROCESSING

A. Dataset Overview

To Build this system we created our own custom Indian Sign *ISL* dataset for model training and evaluation. The dataset contains 720 gesture images representing digits 0 – 9 and alphabets A – Z based on Indian Sign Language gestures.

The dataset characteristics are shown in Table II.

TABLE II
DATASET OVERVIEW

Parameter	Value
Dataset Type	Custom Dataset
Total Images	720
Classes	36
Image Format	JPEG
Input Size	224 × 224
Images per Class	Approximately 20

Images were captured through webcam under various lighting conditions, hand orientations, and backgrounds to improve model robustness.

B. Data Preprocessing

Several preprocessing techniques were applied before training the model.

1) *Image Resizing*: All images were resized to 224 × 224 pixels to maintain consistent input dimensions.

2) *Normalization*: Pixel values were normalized between 0 and 1 to improve training stability.

3) *Landmark Extraction*: MediaPipe Hands was used to extract 21 hand landmarks from each image.

4) *Feature Vector Transformation*: The system transformed the extracted landmark coordinates into a normalized, one-dimensional feature vector, which served as the input for the Random Forest Classifier.

5) *Data Augmentation*: To improve dataset diversity and reduce overfitting, augmentation techniques such as horizontal flipping and brightness adjustments were applied.

C. Exploratory Data Analysis

To investigate how the dataset is distributed and how it varies across the different types of hand gestures, we performed Exploratory Data Analysis (EDA) to confirm this.

The dataset is nearly balanced in terms of the number of samples per class of gesture, thus ensuring a fair, unbiased model to train against. Variations in hand orientation, finger movement, background lighting conditions, and camera distance between gestures were present within the dataset (i.e., gesture images) and add to the robustness and generalization ability of the model for operating in real-time recognition.

The dataset also contained minor background variations but no significant artifacts that could negatively impact training.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Setup

The system's software framework was developed using Python, while its real-time testing was carried out on a standard CPU-based computer integrated with a webcam.

The software stack included:

- Python 3.10
- OpenCV
- MediaPipe
- Scikit-learn
- NumPy
- SymSpellPy
- Google Text-to-Speech gTTS
- Deep Translator
- Playsound
- Pickle
- PyCharm IDE

The dataset was divided into training and testing sets using train-test split methodology.

B. Performance Metrics

The performance of the developed model was evaluated using the following evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Accuracy was calculated :

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

C. Results

The Random Forest classifier achieved an overall accuracy of approximately 99%.

Table III presents the performance metrics.

TABLE III
PERFORMANCE EVALUATION

Metric	Value
Accuracy	99%
Precision	99%
Recall	99%
F1-Score	99%

D. Confusion Analysis

The classifier identified most gestures with high confidence, though minor ambiguities occasionally persisted between visually similar signs.

Some confusion was observed between:

- Alphabet gestures with similar finger positioning.
- Dynamic hand orientations under poor lighting conditions.

To further increase the gesture recognition accuracy and overall system performance, we plan to expand the dataset and incorporate advanced sequence-based learning techniques.

E. Advantages of Proposed System

The proposed system goes beyond basic gesture recognition to serve as a real-time, AI-powered communication assistant for Indian Sign Language (ISL) users. It combines gesture recognition, auto-correct, next-word prediction, multilingual translation, and natural speech generation into a single pipeline. The result is an interactive, highly accurate, and accessible assistive tool for daily communication.

The proposed system offers several advantages:

- Real-time hand gesture recognition via standard webcam input.
- Lightweight and computationally efficient system architecture.
- Zero dependence on specialized hardware or dedicated GPUs.
- Automated spell correction and predictive word suggestions.
- Seamless Odia translation and automated speech generation.
- Low-overhead deployment on standard CPU-based computers.
- Highly affordable and accessible assistive communication tool.
- Improved interaction between hearing-impaired individuals and non-sign-language users.

F. Limitations

Although the developed system provide high performance but several limitations still exist in the current implementation like:

- Limited dataset size

- Difficulty recognizing continuous sentence-level gestures
- Sensitivity to extreme lighting conditions
- Dependence on hand visibility in webcam frames
- It doesnot convert all english word to odia language due to limited dataset.

Future work will focus on addressing these limitations.

VI. PROPOSED SYSTEM OVERVIEW

The initial version of the system was primarily designed for isolated Indian Sign Language (ISL) gesture recognition combined with basic speech conversion. Although such systems can identify individual hand signs, they are Insufficient for real-world communication since they lack contextual intelligence, interactive mechanisms, typing stability, multilingual adaptability, and user assistance features.

To overcome these limitations and enhance usability, accessibility, robustness, and communication efficiency, the developed research extends the conventional gesture recognition framework into an **AI-powered multilingual smart gesture keyboard** capable of intelligent real-time human–computer interaction.

A. System Enhancements

The enhanced system integrates several advanced modules:

- Real-time gesture typing
- MediaPipe-based 3D hand landmark extraction
- Random Forest-based gesture classification
- Gesture stabilization through hold mechanisms
- Prediction history filtering
- SymSpell-based intelligent spell correction
- Next-word prediction
- Multilingual transliteration display
- Bilingual Odia speech synthesis
- Clickable suggestion system
- Interactive UI controls
- Intelligent communication assistance

Unlike traditional gesture recognition systems that Basically classify isolated signs, the proposed framework work as a Complete **AI-powered multilingual assistive communication platform**. It transforms Indian Sign Language (ISL) gestures into meaningful text, transliterated outputs, and speech in real time.

The system unifies computer vision, machine learning, natural language processing (NLP), multilingual transliteration, and speech synthesis into a single intelligent smart gesture keyboard capable of seamless human–computer interaction.

B. Gesture Recognition Pipeline

The system continuously processes live webcam input, extracting 21 hand landmarks that represent finger joints and palm positions through MediaPipe. These spatial coordinates are normalized and converted into feature vectors for classification.

Gesture prediction is performed using a Random Forest classifier, ensuring efficient real-time recognition. To counteract instability caused by temporary hand movement variations, the pipeline integrates prediction history filtering and a gesture-holding mechanism.

The prediction history module stores recent outputs and selects stable predictions across consecutive frames rather than relying on single-frame results. This reduces flickering and enhances classification consistency during continuous capture.

Additionally, the gesture-holding mechanism ensures that characters are appended only when a gesture remains steady for a predefined duration. This approach minimizes noisy predictions, repeated character entries, accidental typing errors, and unstable outputs during real-time typing.

Together, these stabilization strategies improve typing reliability, user control, and communication efficiency, enabling smooth gesture-based interaction without the need for wearable sensors.

C. Intelligent Typing Assistance

Beyond recognition, the system incorporates several NLP-based modules designed to enhance typing quality, communication speed, and user experience:

- **Spell Correction with SymSpell:** A fast dictionary-based correction algorithm automatically resolves errors caused by orientation changes or prediction instability, improving sentence accuracy and readability.
- **Next-Word Prediction:** Context-aware prediction suggests commonly used word sequences, accelerating typing and improving communication efficiency.
- **Clickable Word Suggestions:** Interactive suggestions allow users to select corrected or predicted words directly, reducing manual gesture input effort.
- **Multilingual Transliteration:** Recognized English text is converted into Odia transliteration (e.g., “Namaskara”), supporting regional accessibility and multilingual communication.
- **Bilingual Speech Synthesis:** Text outputs are translated into Odia and converted into real-time speech using Google Text-to-Speech (gTTS), enabling effective audio-based communication.
- **Interactive Smart Keyboard Interface:** Integrated UI controls such as Speak, Clear, and Stop buttons transform the solution into a lightweight AI-powered communication keyboard.

D. Multilingual Communication Support

To improve regional accessibility, the system integrates transliterated Odia text generation and bilingual speech synthesis. Recognized English outputs are converted into Odia transliteration for readability, while Google Text-to-Speech (gTTS) produces real-time Odia audio.

This multilingual capability enables the system to serve not only as a sign language recognition model but also as a lightweight assistive keyboard for hearing-impaired individuals and regional language users.

E. Deployment and Efficiency

The system is optimized to run on standard CPU-based devices without requiring GPUs or costly sensor hardware. Its lightweight architecture, reduced computational complexity, and real-time performance make it affordable, portable, and scalable. This ensures practical deployment across educational institutions, public communication environments, healthcare assistance, and smart human-computer interaction applications.

F. Summary of Features

The major intelligent features integrated into the proposed smart gesture keyboard system are:

- 1) Real-time gesture typing
- 2) MediaPipe-based 3D hand landmark extraction
- 3) Random Forest gesture classification
- 4) Gesture stabilization via hold mechanism
- 5) Prediction history filtering
- 6) SymSpell-based spell correction
- 7) Next-word prediction
- 8) Multilingual transliteration (English to Odia)
- 9) Bilingual Odia speech synthesis
- 10) Clickable word suggestions
- 11) Interactive UI controls
- 12) Intelligent communication assistance

G. Gesture Typing

The system enables users to type text through real-time ISL hand gestures. Each recognized gesture corresponds to a character, digit, or command. This allows hearing-impaired individuals to communicate naturally without requiring a physical keyboard.

H. Character Recognition

The core gesture recognition module identifies ISL alphabets and digits using MediaPipe landmark extraction and Random Forest classification. The recognized characters are continuously appended to form meaningful words and sentences.

I. Spell Correction

Real-time gesture recognition may occasionally produce incorrect characters due to lighting conditions, gesture similarity, or hand orientation variations. To address this issue, an automatic spell correction module is integrated into the system.

The spell correction component compares generated words against a predefined vocabulary and suggests corrected words using edit-distance-based matching techniques.

J. Clickable Word Suggestions

The proposed system provides clickable word suggestions similar to smartphone keyboards. Suggested words are displayed on the user interface, enabling users to quickly select intended words without repeatedly performing gestures.

This feature significantly improves typing speed and reduces user effort.

K. Next Word Prediction

An intelligent next-word prediction module is incorporated to improve sentence completion speed. Based on previously recognized words, the system predicts the most probable next words.

The prediction mechanism improves communication efficiency and enhances user experience.

L. Voice Output Generation

Recognized text is converted into audible speech using a text-to-speech engine. This allows communication between hearing-impaired users and non-sign-language users in real time.

M. Interactive User Interface

The proposed system includes an interactive graphical user interface *GUI* containing multiple control buttons such as:

- STOP Button
- CLEAR Button
- SPEAK Button

The STOP button terminates gesture recognition, the CLEAR button resets generated text, and the SPEAK button converts recognized text into speech.

N. Advantages of the Proposed Smart Keyboard System

The enhanced system offers several practical advantages:

- Real-time ISL communication support
- Faster sentence generation through predictions and suggestions
- Improved typing efficiency
- Reduced communication effort
- Affordable deployment on standard devices
- Improved accessibility for hearing-impaired individuals

O. Drawbacks and Challenges

Although the proposed AI-powered multilingual smart gesture keyboard demonstrates efficient real-time performance and high recognition accuracy, several limitations and open research challenges remain.

- **Challenges in Continuous Dynamic Gesture Recognition:** The system is primarily designed for isolated ISL gestures. Extending recognition to continuous, sentence-level dynamic gestures is difficult due to the temporal complexity of hand movements.
- **Dependence on Lighting Conditions:** Recognition accuracy can degrade in environments with poor illumination, strong shadows, or excessive brightness, as these factors affect the reliability of MediaPipe landmark detection.
- **Confusion from Gesture Similarity:** Certain ISL signs share similar finger orientations and hand postures, which may lead to misclassification or ambiguity in prediction.
- **Restricted Dataset Size:** The dataset used for training is relatively small and lacks diversity. Variations in hand shape, skin tone, gesture speed, camera angle, and signing style can reduce the model's generalization ability.

- **Reliance on Clear Hand Visibility:** Accurate recognition requires unobstructed visibility of the hand within the webcam frame. Occlusion, motion blur, or improper positioning can negatively impact prediction quality.
- **Residual Prediction Instability:** Despite stabilization mechanisms such as gesture holding and history filtering, occasional instability may still produce noisy outputs during continuous capture.
- **Vocabulary Constraints in NLP Modules:** The SymSpell correction and next-word prediction modules currently operate on limited vocabulary sets. Larger datasets and advanced language models are needed for richer, context-aware communication.
- **Regional Variability in ISL:** Indian Sign Language exhibits regional differences and user-specific variations. These inconsistencies can affect recognition accuracy and system adaptability.
- **Impact of Background Clutter:** Complex or dynamic backgrounds may interfere with landmark extraction, reducing robustness in uncontrolled environments.
- **Hardware and Webcam Limitations:** Low-resolution cameras, reduced frame rates, or limited CPU resources can introduce latency and hinder real-time performance.
- **Restricted Multilingual Capabilities:** At present, the system supports English text and Odia transliterated speech. Expanding to additional Indian languages remains an important future direction.
- **Absence of Advanced Deep Learning Models:** The current implementation relies on a lightweight Random Forest classifier for efficiency on CPUs. Incorporating transformer-based or deep learning architectures could further enhance recognition accuracy, particularly for large-scale dynamic gesture datasets.

VII. FUTURE SCOPE

The proposed system remains adaptable to further modifications aimed at enhancing its overall accuracy, environmental robustness, and communication efficiency.

It includes:

- Expansion of the ISL gesture dataset to support larger vocabulary and sentence-level communication
- Integration of dynamic gesture recognition for continuous real-time sign interpretation
- Addition of multilingual translation and speech synthesis support for wider accessibility
- Incorporation of facial expression and body pose analysis for improved contextual understanding
- Integration of advanced deep learning models such as LSTM, GRU, and Transformer architectures for sequential gesture prediction
- Improvement of recognition performance under varying lighting conditions and complex backgrounds
- Development of offline translation and speech generation modules to reduce internet dependency

These improvements can significantly enhance system performance and usability.

VIII. CONCLUSION

This system made for real-time Indian Sign Language recognition. It utilize a multi-language intelligent gesture keyboard that are developed using MediaPipe and machine-learning techniques.

This system collects live hand gestures through a webcam, extracts 21 distinct hand-joint locations via MediaPipe, normalizes them into shared coordinates, and classifies the gestures using Random Forest to achieve on-the-fly prediction.

Unlike typical isolated gesture recognition methods, the proposed intelligent gesture keyboard serves as a holistic AI-assisted communication platform for Indian Sign Language users by enabling real-time gesturing and typing, intelligent text generation from gestures, multilingual transliteration from Indian Sign Language to target languages (e.g., English, Hindi, etc.), and bilingual speech synthesis from Indian Sign Language to audio output.

The proposed intelligent keyboard system includes several specialized modules: gesture stabilization using a gesture-holding mechanism, prediction history filtering, SymSpell spelling correction, next-word prediction, word suggestions with pointer controls, a clickable keyboard, Odia transliteration, and bilingual speech synthesis using Google Text-to-Speech (gTTS).

By leveraging landmark-based image recognition, the intelligent gesture keyboard reduces the computational burden (eliminating the requirement for GPUs to process images or gestures) for standard CPU-based applications while achieving high accuracy and low latency.

Real-world experiments show that the proposed multi-language intelligent gesture keyboard system supports stable and reliable real-time typing performance, improved typing accuracy rates, minimized false predictions, and enhanced ease of use via natural language processing (NLP) during user text input.

We designed this affordable, lightweight, and scalable multilingual solution to serve as an assistive communication device for hard-of-hearing individuals and regional language speakers. The system effectively bridges the gap between sign language communication and everyday public interactions.

Moving forward, our research will explore continuous sentence-level ISL translation, integrate transformer models for language generation, and apply deep learning to dynamic gesture recognition. We also plan to incorporate more Indian languages and develop cloud, mobile, and web applications to scale the platform for global accessibility.

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